Research, Wireframe, Test Cases

Feature 1: User Data and Storage

Research & Design:

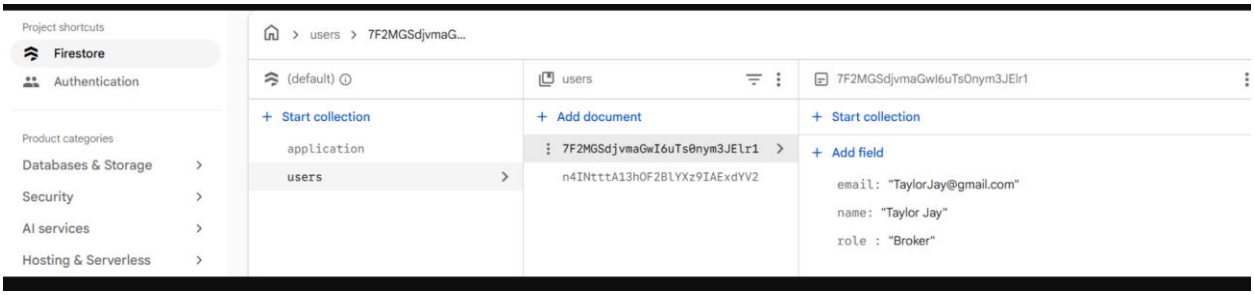
Storing data is essential, I went for a simple approach initially where a user inputs anything and it gets put into simple database table, check simple-data-storage-idea but it is too basic and only gets stored locally. So, after consulting with the one responsible for AL-Core and Ledger, we decided to use Firebase, Googles Backend-as-a-Service (BaaS) platform, as our database.

I researched ways for connectivity between my web and the Firebase and stumbled across code on gitea: https://gitea.com/saad2134/donorsync/src/branch/main/firebaseConfig.ts it has a config to connect to Firebase via apiKey, authDomain, projectId, storageBucket, messagingSenderId, and appId.

```
export const firebaseConfig = {
  apiKey: "AIzaSyD_ZAWsAh1cu1Z_jOxrDTmUooxbX2YbpR4",
  authDomain: "agentic-lender-668fa.firebaseapp.com",
  projectId: "agentic-lender-668fa",
  storageBucket: "agentic-lender-668fa.firebasestorage.app",
  messagingSenderId: "333912556534",
  appId: "1:333912556534:web:54ed7c417c33add1d6fdd7"
};
```

I also had to configure Firebase rules so it would give less restrictions; in firestore.rules.

In the final product for now, I have a broker or underwriter to sign in with name, email and password, and so the firebase stores their role, full name, email, and password.



^^^^ The Firebase allowing fields for web and example

Wireframe / Sketch:

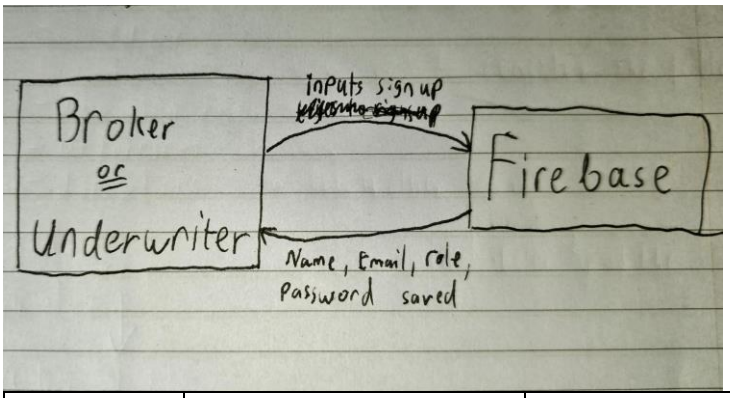
Test Case:

If broker or an Underwriter can login, then it means that the Firebase is connected. I did personal tests and I made an ai test its connectivity.

Feature 1 – Live test against Firebase project agentic-lender-668fa

25/10/2026, 13:19:06 · test user al-web-test-1792934342945@example.com · test login deleted

ID	Test	Expected	Actual	Result
TC-03	Read users without logging in	Blocked	Blocked (403)	PASS
TC-01	Create account (Firebase Authentication)	Account created	Created, uid nGrLOrR4...	PASS
TC-01	Save email, name, role to Firestore users/{uid}	Saved	Saved	PASS
TC-02	Fields stored in Firestore (password must NOT be there)	email, name, role only	email, name, role	PASS
TC-02	Log in with wrong password	Rejected	Rejected (INVALID_LOGIN_CREDENTIALS)	PASS
TC-04	Broker changes own role to Underwriter	Blocked	Blocked (403)	PASS
TC-04	Broker reads ALL applications	Blocked	Blocked (403)	PASS
TC-04	Broker reads applications where brokerID = self	Allowed	Allowed	PASS



Test ID	Story	Steps	Expected
US1-T01	Dev: a place to store user data	Sign up a test broker	Account in Firebase Auth; users/{uid} saved with email, name, role

US1-T02	Data stored securely	Check the saved fields; log in with a wrong password	No password in Firebase; wrong password rejected
US1-T03	Data stored securely	Read users without logging in	Blocked
US1-T04	Data stored securely	As a broker: change own role; read all applications; read own applications	Blocked; blocked; allowed

Feature 2: Authentication and role-based access

Research & Design:

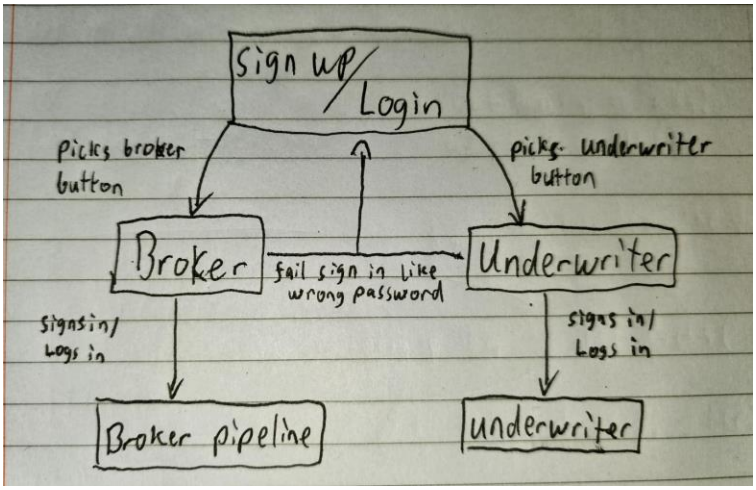
As the backlog suggests, I want to make role-based access so that a user picks if they are a broker or underwriter when signing up and logging in. If you are broker you see your client files and other things, while the bank underwriter has portfolio-wide oversite. So we need to separate them.

Initially I thought on using a server. While I searched for code that could help I found:
https://github.com/SoheelGadad/Userpage_nodejs
https://github.com/subomi/loginportal

They used a server locally to restrict or give user roles. But the only issue is I figured that using Firebase would be better in the long run instead of running a terminal or server every time. But still these repositories were a great foreground.

In the final product I have a working sign up & login where the user selects a button to confirm if they are broker or underwriter, and depending on what they picked -> they sign up -> full name, email, password, role gets saved in firebase -> broker sees what broker sees; alternatively, underwriter sees what only underwriter should see. Other things include putting wrong information like email, password for sign up/login, gets rejected and restarts sign up/login.

Wireframe / Sketch:



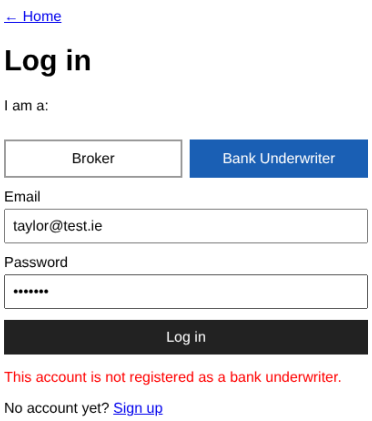
Test Case:

Test ID	Story	Steps	Expected
US2-T01	Any user	Open the site	Home page with Sign up and Log in
US2-T02	Sign up	Fill in the form without picking a role, then submit	“Choose Broker or Bank Underwriter.”
US2-T03	Sign up as broker	Pick Broker, fill in the form, Create account	Lands on Broker Pipeline
US2-T04	Redirect and log out	Click Log out, then type broker.html	Back to Home; then redirected to login
US2-T05	Broker logs in	Pick Broker, correct email and password	Broker Pipeline shows the broker’s email
US2-T06	Broker logs in	Correct email, wrong password	“Wrong email or password.”
US2-T07	Role-based access	Broker account with Bank Underwriter selected	“This account is not registered as a bank underwriter.”
US2-T08	Role-based access	Logged-in broker types underwriter.html	Redirected to login
US2-T09	Underwriter logs in	Pick Bank Underwriter, correct details	Underwriter Pipeline (portfolio-wide)

US2-T03 Pick Broker, fill in the form, Create account



US2-T07 Broker account with Bank Underwriter selected



Feature 3: Origination pipeline view

Research & Design:

As I already have the role base access, the question is what broker or an underwriter sees in there pipeline after sign up. I think we have already determined previously but to go over again.

Broker: restricted to viewing and managing only their referred client files displaying incoming loan applications, categorized by LTV thresholds, risk tiers.

Bank Underwriter: portfolio-wide oversight with sign-off authority displaying incoming loan applications, categorized by LTV thresholds, risk tiers, and lifecycle status (Applied, AI Deliberated, Underwriter Approved, Funded, Closed).

For the final design I have what is above, but actions are restricted for the moment

Wireframe / Sketch:

Figma designs:

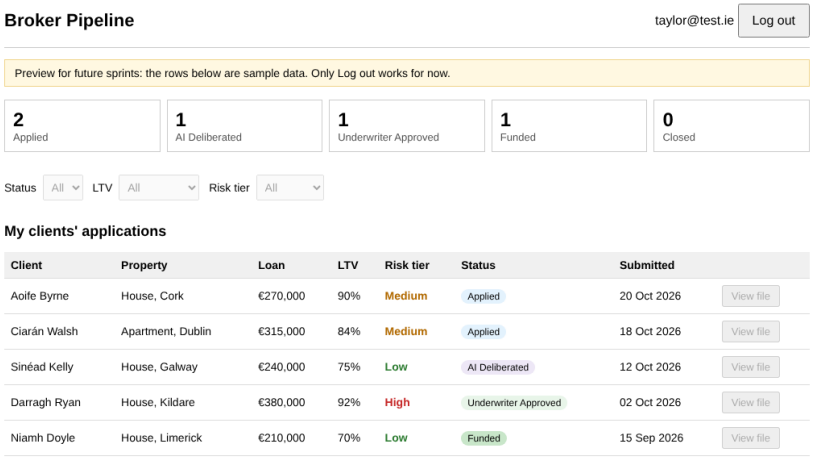




Test Case:

Test ID	Story	Steps	Expected
US3-T01	Broker sees own files	Log in as broker	Table of own clients with status, LTV, risk tier; status counts
US3-T02	Broker filters by status and LTV	Try the Status and LTV filters	List narrows to matching files
US3-T03	Underwriter sees all applications	Log in as underwriter	All applications with status, LTV, risk tier, broker; LTV and risk summary

Broker pipeline



Bank Underwriter pipeline

